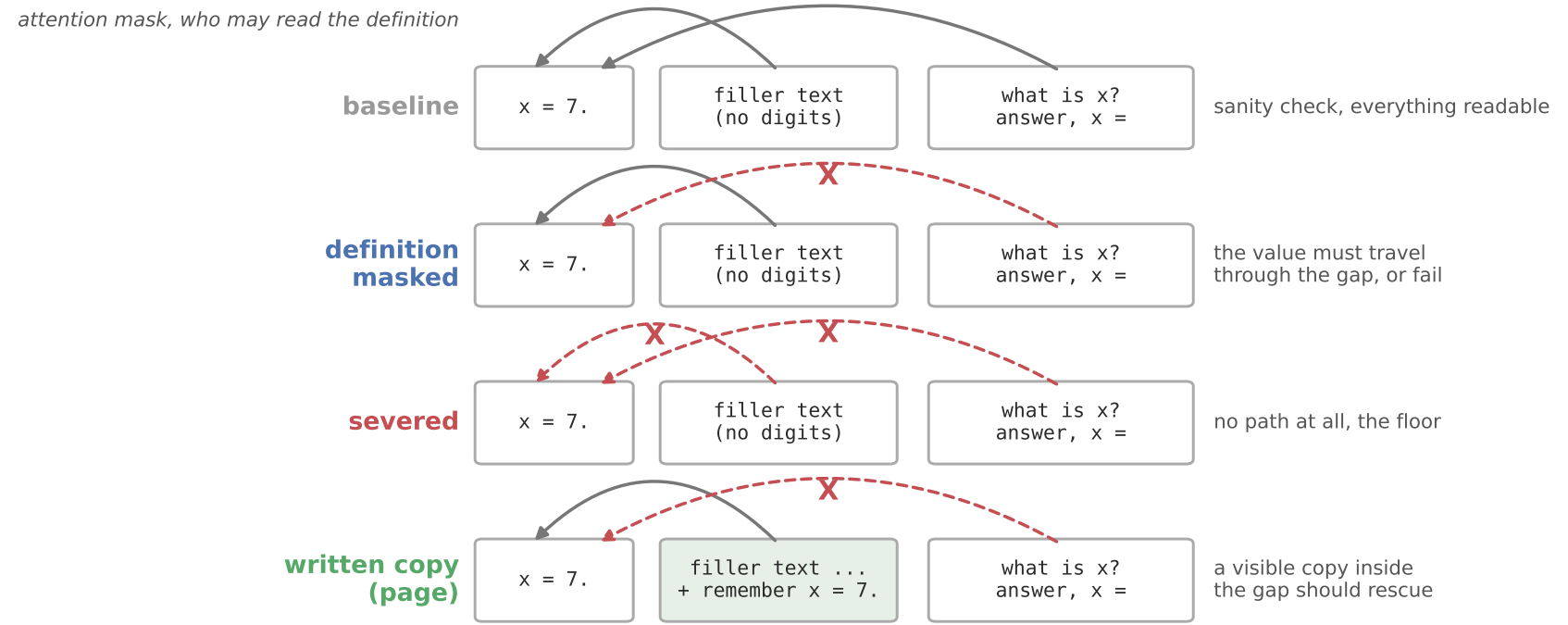The masked-maintenance design